AP Calculus AB Module 5B
First Derivative Analysis

Unit 5: Analytical Applications of Differentiation, Sections 5.3–5.5
Topics: where a function is increasing or decreasing from the sign of f'; the First Derivative Test for relative (local) extrema; the Candidates Test for absolute (global) extrema on a closed interval

Learning Goals

By the end of this module, you should be able to:

- find the **critical points** of a function (where $f'(x) = 0$ or $f'(x)$ does not exist) and explain why those are the only places where the behavior of f can change;
- determine the intervals on which a function is **increasing or decreasing** from the sign of its first derivative, and *justify* the answer in the language an AP reader expects;
- apply the **First Derivative Test** to classify each critical point as a relative maximum, a relative minimum, or neither;
- apply the **Candidates Test** to locate the *absolute* (global) maximum and minimum of a continuous function on a closed interval;
- read all of this information *backwards* from a **graph of** f' (or a sign chart / table), distinguishing the behavior of f from the behavior of f';
- state every conclusion *in context*, with correct units, and with a complete justification rather than a bare answer.

Note on the 2026 CED (Unit 5). The Extreme Value Theorem is stated this way for Fall 2026: if a function f is continuous over the closed interval $[a, b]$, then f has *at least one minimum value and at least one maximum value* on $[a, b]$. That guarantee is exactly what makes the **Candidates Test** valid: on a closed interval a continuous function *must* attain an absolute max and an absolute min, and they can only occur at critical points or endpoints. The EVT itself is taught in Module 5A; here we use it.

Condensed Lecture

This module turns one fact, that the sign of f' tells you which way f is going, into three skills: finding where f rises and falls, locating its local peaks and valleys, and finding its highest and lowest points on a closed interval. Here is the whole toolkit in brief; the Full Lecture explains the *why* behind each piece.

Increasing, decreasing, and critical points

The first derivative $f'(x)$ is the slope of f. Its sign controls the direction of f.

If $f'(x) > 0$ on an interval, then f is **increasing** there.
If $f'(x) < 0$ on an interval, then f is **decreasing** there.
A **critical point** is an x-value in the domain of f where

$$f'(x) = 0 \qquad \text{or} \qquad f'(x) \text{ does not exist.}$$

These are the only places where f can switch between increasing and decreasing.

The sign chart (the core procedure)

Almost every problem runs the same routine: compute f' and **factor it**; find the critical points; mark them on a number line; test one point in each interval to get the sign of f'; then label each interval increasing ($\nearrow$, where $f' > 0$) or decreasing ($\searrow$, where $f' < 0$).

First Derivative Test: relative extrema (works anywhere)

Read the relative (local) maxima and minima straight off the sign chart.

At a critical point $x = c$:

- if f' changes from positive to negative, then f has a **relative maximum** at $x = c$;
- if f' changes from negative to positive, then f has a **relative minimum** at $x = c$;
- if f' does not change sign, then f has **neither** at $x = c$.

A relative extremum can occur wherever a sign change happens, but never at an endpoint of a closed interval.

Candidates Test: absolute extrema (closed interval only)

When a question asks for the *absolute* (global) max or min of a continuous f on $[a, b]$: evaluate f at every critical point in (a, b) **and** at both endpoints a, b; the largest of those outputs is the absolute maximum and the smallest is the absolute minimum. You compare y-values, and you must include the endpoints.

Reading a graph of f'

If you are handed the graph of f' (not f), read its *height*: where the f' graph is above the x-axis, f is increasing; below the axis, f is decreasing; where it crosses the axis, f has a critical point (and the way it crosses gives the max/min). A point where the graph only *touches* the axis is a critical point with no sign change, which is neither a max nor a min.

Justifying your answer

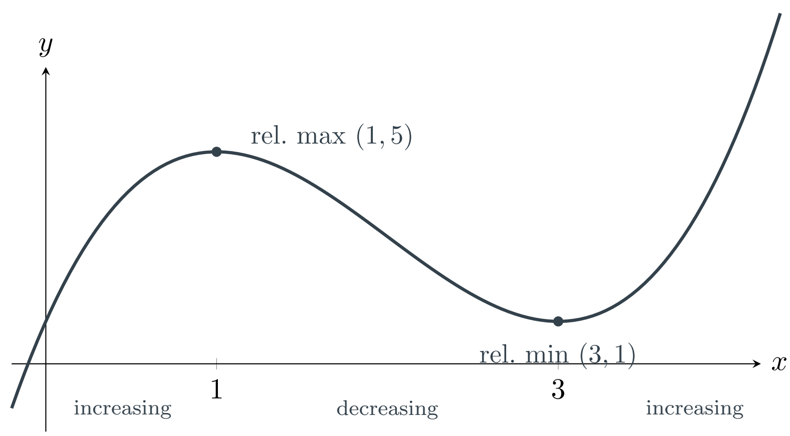
On the AP Exam a correct interval or x-value earns full credit only when it is justified by the behavior of f'. Every conclusion in this module ends by pointing to the sign of f' or to a sign change in f' (templates in the Full Lecture).

Full Lecture

Why the sign of the derivative controls the shape of the graph

Picture walking left to right along the graph of f. When the slope under your feet is positive you are walking *uphill*, and the outputs are getting larger, so f is increasing. When the slope is negative you are walking *downhill*, so f is decreasing. The places where the hill changes direction (the tops of peaks, the bottoms of valleys) are exactly where the slope passes through zero or hits a sudden corner. That is the whole story, and it is worth seeing once on a real graph.

Take $f(x) = x^3 - 6x^2 + 9x + 1$. The graph climbs, levels off at a peak, falls into a valley, then climbs again. The labels below the axis name what the slope is doing on each stretch.



The peak at $x = 1$ and the valley at $x = 3$ are where the slope is momentarily zero, where the tangent line is horizontal. Between them the curve falls, so f' is negative; outside them it rises, so f' is positive. We now turn that picture into a reliable procedure.

Critical points: the only places behavior can change

If f is increasing right up to some point and decreasing right after it, the slope had to pass from positive to negative. For a function whose derivative exists everywhere, that means the slope passed *through zero*: $f'(c) = 0$. But the slope can also flip without ever equaling zero, at a sharp **corner** or **cusp**, where f' simply does not exist. Both cases are captured by one definition.

$x = c$ is a **critical point** of f if c is in the domain of f and either

$$f'(c) = 0 \qquad \text{or} \qquad f'(c) \text{ does not exist.}$$

Increasing/decreasing behavior can only change at a critical point, so these are the only candidates for relative extrema.

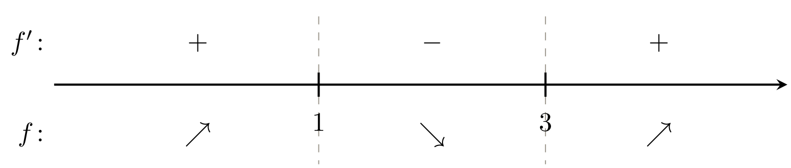
Trap: a critical point is *not* automatically a max or a min. Having $f'(c) = 0$ only means the graph is momentarily flat there. The function might keep going the same direction. Think of $f(x) = x^3$ at $x = 0$: the slope is zero, but f is increasing on both sides, so $x = 0$ is neither a max nor a min. You must check whether f' *changes sign*, not just whether it equals zero.

The sign chart: your main tool

A **sign chart** for f' organizes everything. The recipe is always the same:

1. Find $f'(x)$ and write it in *factored* form.
2. Find every critical point (set each factor equal to zero; also note where f' is undefined).
3. Mark those critical points on a number line; they cut the line into intervals.
4. In each interval, test the *sign* of f' by plugging one convenient point into the factored derivative.
5. Read off: $f' > 0$ means f is increasing ($\nearrow$); $f' < 0$ means f is decreasing ($\searrow$).

For $f(x) = x^3 - 6x^2 + 9x + 1$ we have $f'(x) = 3x^2 - 12x + 9 = 3(x-1)(x-3)$, with critical points $x = 1$ and $x = 3$. Testing one point in each interval gives this chart:



Reading the chart: f is increasing on $(-\infty, 1)$ and on $(3, \infty)$, and decreasing on $(1, 3)$. At $x = 1$ the slope flips from $+$ to $-$ (a peak); at $x = 3$ it flips from $-$ to $+$ (a valley).

The First Derivative Test (relative extrema)

A **relative** (or **local**) **maximum** is a point higher than everything immediately around it: the top of a local hill, not necessarily the highest point on the whole graph. A relative minimum is the bottom of a local valley. The First Derivative Test reads these straight off the sign chart.

First Derivative Test. Suppose c is a critical point of a continuous function f.

- If f' changes from **positive to negative** at c, then f has a **relative maximum** at $x = c$.
- If f' changes from **negative to positive** at c, then f has a **relative minimum** at $x = c$.
- If f' does **not change sign** at c, then f has **neither** at $x = c$.

Trap: relative extrema never occur at the endpoints of a closed interval. "Relative" means the point beats the values on *both* sides of it, but an endpoint has values on only one side. Endpoints can still be *absolute* extrema (next section), just not relative ones. AP readers watch for this distinction closely.

From relative to absolute: the Candidates Test

The First Derivative Test finds local peaks and valleys, but it cannot tell you which point is the *highest* or *lowest* on a closed interval, since an endpoint might beat every interior peak. When the question asks for the **absolute** (global) maximum or minimum on a closed interval $[a, b]$, use the Candidates Test.

Why it works: because f is continuous on the closed interval, the Extreme Value Theorem guarantees that an absolute max and an absolute min both exist. An absolute extremum is either an interior point (where it must also be a local extremum, so $f' = 0$ or DNE, i.e. a critical point) or it sits at an endpoint. So the complete list of "candidates" is short: the critical points inside the interval, plus the two endpoints.

Candidates Test (absolute extrema on $[a, b]$). For f continuous on $[a, b]$:

1. List every critical point in the open interval (a, b).
2. Evaluate f at each of those critical points *and* at both endpoints a and b.
3. The largest output is the absolute maximum; the smallest output is the absolute minimum.

You are comparing y-values (outputs of f), not x-values, and you must include the endpoints.

Tip: organize the Candidates Test as a tiny table. List each candidate x in one column and its $f(x)$ in the next. The biggest and smallest $f(x)$ jump out, and the table is itself the "work" an AP reader wants to see.

Reading it all backwards from a graph of f'

A very common and very point-rich AP question gives you the graph of f' (the derivative) and asks about f (the original function). The key is to keep straight *which* graph you are looking at: the height of the f' graph is the *slope* of f.

How to read a graph of f':

- f' graph *above* the x-axis ($f' > 0$): f is **increasing**.
- f' graph *below* the x-axis ($f' < 0$): f is **decreasing**.
- f' graph *crosses* the x-axis: a critical point of f; the way it crosses (above-to-below or below-to-above) gives the relative max/min.
- f' graph *touches* the axis without crossing: a critical point, but with no sign change, so neither a max nor a min.

Trap: don't confuse "f' is increasing" with "f is increasing." If the f' graph is sloping upward but still *below* the axis, then $f' < 0$, so f is *decreasing* even though f' is increasing. Always ask about the *sign* (height) of f', not the direction the f' graph is moving.

Writing the answer (AP language)

On the AP Exam, a correct interval or x-value earns full credit only when it is *justified*. "Justify" means you must point to the behavior of f'. The templates below are the exact sentences readers reward.

Increasing/decreasing: "f is *(increasing/decreasing)* on *(interval)* because $f'(x)$ is *(positive/negative)* on that interval."

First Derivative Test: "f has a relative *(maximum/minimum)* at $x = c$ because f' changes from *(positive to negative / negative to positive)* at $x = c$."

Neither: "$x = c$ is a critical point, but f' does *not* change sign there, so f has neither a relative maximum nor a relative minimum at $x = c$."

Candidates Test: "Because f is continuous on $[a, b]$, the absolute *(maximum/minimum)* of f is *(value)* at $x =$*(location)*, found by comparing the values of f at the critical points and the endpoints."

Using the calculator (when the exam allows it)

On calculator-active questions, the AP exam expects you to use only four built-in operations: (1) graph a function, (2) find the zero (root) or intersection of graphs, (3) compute the numerical derivative at a point, and (4) compute a definite integral. For first-derivative analysis the useful ones are:

- **Finding critical points.** If $f'(x)$ is messy to factor, graph $y = f'(x)$ and use the calculator's *zero* feature to find where $f'(x) = 0$. Report those x-values.
- **Evaluating candidates.** To run the Candidates Test, store the function as Y_1 and evaluate Y_1 at each candidate x (critical points and endpoints), then compare.

Calculator traps. (1) On a calculator-active free-response problem you must still write down what you found, e.g. "$f'(x) = 0$ at $x \approx 1.272$", and your reasoning; a bare calculator answer earns little. (2) On a *no-calculator* question you must show the algebra (factoring f', the sign chart) by hand. Know which section you are in.

Model Problem 1: Increasing/Decreasing and the First Derivative Test

Let $f(x) = x^3 - 6x^2 + 9x + 1$.

(a) Find all critical points of f.

(b) Determine the open intervals on which f is increasing and on which it is decreasing. Justify your answer.

(c) Classify each critical point as a relative maximum, relative minimum, or neither, and find the value of f there.

Solution

(a) Critical points.

Differentiate term by term with the power rule, then factor completely:

$$f'(x) = 3x^2 - 12x + 9 = 3\big(x^2 - 4x + 3\big) = 3(x-1)(x-3).$$

Because f' is a polynomial, it exists for every x, so the only critical points come from setting it equal to zero:

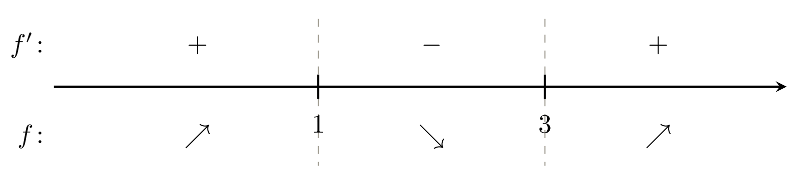
$$3(x-1)(x-3) = 0 \implies x = 1 \quad \text{or} \quad x = 3.$$

The critical points are $x = 1$ and $x = 3$.

(b) Where f is increasing and decreasing.

The two critical points from part (a) split the number line into three intervals. Using the factored derivative $f'(x) = 3(x-1)(x-3)$, test one convenient point in each:

$$f'(0) = 3(-1)(-3) = +9 > 0, \qquad f'(2) = 3(1)(-1) = -3 < 0, \qquad f'(4) = 3(3)(1) = +9 > 0.$$



So f is **increasing on** $(-\infty, 1)$ **and** $(3, \infty)$, because $f'(x) > 0$ there, and **decreasing on** $(1, 3)$, because $f'(x) < 0$ there.

(c) Classifying the critical points.

Reading the sign chart from part (b): at $x = 1$, f' changes from positive to negative, so f has a **relative maximum** there; at $x = 3$, f' changes from negative to positive, so f has a **relative minimum** there. Evaluating f at each:

$$f(1) = 1 - 6 + 9 + 1 = 5, \qquad f(3) = 27 - 54 + 27 + 1 = 1.$$

So f has a relative maximum value of 5 at $x = 1$ and a relative minimum value of 1 at $x = 3$.

Model Problem 2: The Candidates Test (absolute extrema)

Find the absolute maximum and absolute minimum values of $f(x) = x^3 - 3x^2 - 9x + 5$ on the closed interval $[-2, 4]$, and state where each occurs.

Solution

Because f is a polynomial it is continuous on the closed interval $[-2, 4]$, so the Extreme Value Theorem guarantees an absolute maximum and minimum, and the Candidates Test will locate them: they can occur only at critical points inside the interval or at the endpoints.

First find the critical points. Differentiating and factoring,

$$f'(x) = 3x^2 - 6x - 9 = 3\big(x^2 - 2x - 3\big) = 3(x-3)(x+1),$$

so $f'(x) = 0$ at $x = -1$ and $x = 3$, both of which lie inside $(-2, 4)$. The candidates are therefore these two critical points together with the endpoints: $x = -2, -1, 3, 4$.

Evaluating f at each candidate, showing every substitution,

$$\begin{aligned} f(-2) &= (-2)^3 - 3(-2)^2 - 9(-2) + 5 = -8 - 12 + 18 + 5 = 3, \\ f(-1) &= (-1)^3 - 3(-1)^2 - 9(-1) + 5 = -1 - 3 + 9 + 5 = 10, \\ f(3) &= (3)^3 - 3(3)^2 - 9(3) + 5 = 27 - 27 - 27 + 5 = -22, \\ f(4) &= (4)^3 - 3(4)^2 - 9(4) + 5 = 64 - 48 - 36 + 5 = -15. \end{aligned}$$

Collecting these:

x	-2	-1	3	4
$f(x)$	3	10	-22	-15

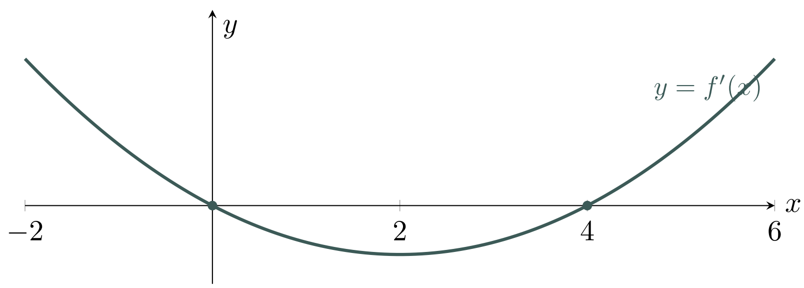
Comparing the outputs, the largest is 10 and the smallest is -22. So, because f is continuous on $[-2, 4]$, the **absolute maximum value is** 10**, at** $x = -1$, and the **absolute minimum value is** -22**, at** $x = 3$, found by comparing the values of f at the critical points and the endpoints.

Model Problem 3: Reasoning from a Graph of f'

The graph below shows $y = f'(x)$, the *derivative* of a function f that is continuous on $[-2, 6]$.

(a) On what open intervals is f increasing? Decreasing? Justify your answers.

(b) Find the x-coordinates of all relative extrema of f, and classify each. Justify.



Solution

(a) Where f is increasing and decreasing.

> Read the *height* of the f' graph, since that height is the slope of f. The graph is above the x-axis on $(-2, 0)$, below it on $(0, 4)$, and above it again on $(4, 6)$. So $f' > 0$ on $(-2, 0)$ and $(4, 6)$, and $f' < 0$ on $(0, 4)$. Therefore f is **increasing on** $(-2, 0)$ **and** $(4, 6)$, because $f'(x) > 0$ there, and **decreasing on** $(0, 4)$, because $f'(x) < 0$ there.

(b) Relative extrema.

> The critical points are where the graph crosses the axis: $x = 0$ and $x = 4$. At $x = 0$ the graph passes from above the axis to below it, so f' changes from positive to negative, giving a **relative maximum** at $x = 0$. At $x = 4$ the graph passes from below to above, so f' changes from negative to positive, giving a **relative minimum** at $x = 4$. (Notice we never needed a formula for f; the sign of f' was enough.)

Guided Problem Solving

Guided 1: Reading increasing/decreasing from a factored derivative

A function f has derivative $f'(x) = (x + 2)(x - 4)$. On which interval is f decreasing?

(A) $(-\infty, -2)$

(B) $(-2, 4)$

(C) $(4, \infty)$

(D) $(-\infty, -2) \cup (4, \infty)$

(E) f is never decreasing.

Solution. f is decreasing where $f'(x) < 0$. The product $(x+2)(x-4)$ is negative between its roots -2 and 4 (one factor positive, the other negative). Outside that interval both factors share a sign, so the product is positive. Thus f is decreasing on $(-2, 4)$. The answer is **(B)**.

Guided 2: Telling a real extremum from a "shelf"

A function f has derivative $f'(x) = x^2(x-3)$. How many relative extrema does f have?

(A) 0

(B) 1

(C) 2

(D) 3

(E) Cannot be determined.

Solution. The critical points are $x = 0$ and $x = 3$. The factor x^2 is never negative, so the sign of $f'(x) = x^2(x-3)$ is controlled by $(x-3)$: negative for $x < 0$ and for $0 < x < 3$, then positive for $x > 3$. At $x = 0$ there is *no sign change* (negative on both sides), so $x = 0$ is neither, a "shelf." At $x = 3$, f' changes from negative to positive, a relative minimum. So there is exactly one relative extremum. The answer is **(B)**.

Guided 3 (FRQ): Full analysis when the derivative needs the product rule

Let $f(x) = x^2 e^{-x}$.

(a) Find $f'(x)$ and write it in fully factored form.

(b) Find the intervals on which f is increasing and decreasing. Justify.

(c) Classify each critical point and give the value of f there.

Solution.

(a) The derivative, factored.

> Use the product rule with $u = x^2$ (so $u' = 2x$) and $v = e^{-x}$ (so $v' = -e^{-x}$, by the chain rule on e^{-x}):
> $$f'(x) = u'v + uv' = 2x\,e^{-x} + x^2\big(-e^{-x}\big) = 2x\,e^{-x} - x^2 e^{-x}.$$
> Factor out the common e^{-x}, then the common x:
> $$f'(x) = e^{-x}\big(2x - x^2\big) = e^{-x}\,x\,(2-x).$$

(b) Where f is increasing and decreasing.

> Since e^{-x} is never zero, the critical points come from $x(2-x) = 0$ [using f' from part (a)], giving $x = 0$ and $x = 2$; f' exists everywhere, so there are no others. Because $e^{-x} > 0$ for all x, the sign of f' equals the sign of $x(2-x)$:
>
> $$x < 0: \ (-)(+) < 0; \qquad 0 < x < 2: \ (+)(+) > 0; \qquad x > 2: \ (+)(-) < 0.$$
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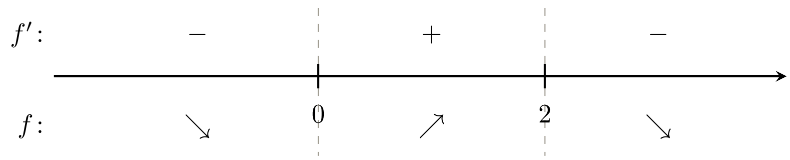
> 
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> So f is increasing on $(0, 2)$, because $f' > 0$ there, and decreasing on $(-\infty, 0)$ and $(2, \infty)$, because $f' < 0$ there.

(c) Classifying the critical points.

> From the sign chart in part (b): at $x = 0$, f' changes from negative to positive, a relative minimum; at $x = 2$, f' changes from positive to negative, a relative maximum. Evaluating,
>
> $$f(0) = 0^2 e^0 = 0, \qquad f(2) = 2^2 e^{-2} = 4e^{-2} \approx 0.541.$$
>
> So f has a relative minimum value of 0 at $x = 0$ and a relative maximum value of $4e^{-2} \approx 0.541$ at $x = 2$.

Guided 4 (FRQ): Candidates Test on a closed interval

Find the absolute maximum and absolute minimum values of $f(x) = 2x^3 + 3x^2 - 12x$ on $[-3, 2]$, and state where each occurs.

Solution. f is a polynomial, hence continuous on the closed interval $[-3, 2]$, so the Candidates Test applies. Differentiating and factoring,

$$f'(x) = 6x^2 + 6x - 12 = 6\big(x^2 + x - 2\big) = 6(x+2)(x-1),$$

so the critical points are $x = -2$ and $x = 1$, both inside $(-3, 2)$. The candidates are these together with the endpoints: $x = -3$, -2, 1, 2. Evaluating f at each,

$$\begin{aligned}
f(-3) &= 2(-27) + 3(9) - 12(-3) = -54 + 27 + 36 = 9,\\
f(-2) &= 2(-8) + 3(4) - 12(-2) = -16 + 12 + 24 = 20,\\
f(1) &= 2(1) + 3(1) - 12(1) = 2 + 3 - 12 = -7,\\
f(2) &= 2(8) + 3(4) - 12(2) = 16 + 12 - 24 = 4.
\end{aligned}$$

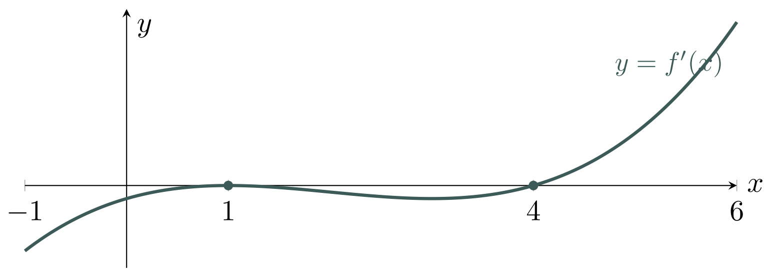
Comparing the four outputs 9, 20, -7, 4: because f is continuous on $[-3, 2]$, the absolute maximum value is 20, at $x = -2$, and the absolute minimum value is -7, at $x = 1$.

Guided 5 (FRQ): A graph of f' that touches without crossing

The graph below shows $y = f'(x)$, the derivative of a function f continuous on $[-1, 6]$. The graph touches the x-axis at $x = 1$ and crosses it at $x = 4$.

(a) On what intervals is f increasing? Decreasing? Justify.

(b) Identify and classify all relative extrema of f. Justify, paying attention to $x = 1$.



Solution.

(a) Where f is increasing and decreasing.

> Read the height of the f' graph. It sits below the axis on $(-1, 1)$ and again on $(1, 4)$ (it only *touches* the axis at $x = 1$, then dips back down), and above the axis on $(4, 6)$. So $f' < 0$ across $(-1, 4)$ (reaching 0 only at the instant $x = 1$) and $f' > 0$ on $(4, 6)$. Therefore f is decreasing on $(-1, 4)$, because $f'(x) < 0$ there (the single point where $f' = 0$ does not interrupt the decrease), and increasing on $(4, 6)$, because $f'(x) > 0$ there.

(b) Relative extrema.

> The critical points are $x = 1$ and $x = 4$. At $x = 4$, f' changes from negative to positive, so f has a **relative minimum** there. At $x = 1$, f' is negative on both sides (it touches zero but does *not* change sign), so $x = 1$ is **neither** a maximum nor a minimum. The only relative extremum is the relative minimum at $x = 4$.

Guided 6: A critical point where the derivative does not exist

Let $f(x) = (x - 1)^{2/3}$. Which best describes f at $x = 1$?

(A) Relative maximum

(B) Relative minimum

(C) Neither; $f'(1) = 0$ with no sign change

(D) No critical point exists at $x = 1$

(E) f is increasing through $x = 1$

Solution. With $f(x) = (x-1)^{2/3}$, differentiate using the power rule:

$$f'(x) = \frac{2}{3}(x-1)^{-1/3} = \frac{2}{3(x-1)^{1/3}}.$$

At $x = 1$ the denominator is 0, so $f'(1)$ does *not* exist, but $x = 1$ is in the domain of f (with $f(1) = 0$), so it *is* a critical point. Checking the sign on each side: for $x < 1$, $(x-1)^{1/3} < 0$ so $f' < 0$ (decreasing); for $x > 1$, $(x-1)^{1/3} > 0$ so $f' > 0$ (increasing). Since f' changes from negative to positive, f has a relative minimum at $x = 1$ (a cusp). The answer is **(B)**.

Guided 7: Why $f'(c) = 0$ is not enough

A student computes $f'(2) = 0$ and immediately concludes that f has a relative maximum at $x = 2$. Which statement is correct?

(A) The student is correct: $f'(2) = 0$ guarantees a relative maximum.

(B) The student is correct only if $f''(2) < 0$, which is automatic.

(C) $x = 2$ is a critical point, but f' must *change from positive to negative* there for a relative maximum; $f'(2) = 0$ alone is not enough.

(D) $f'(2) = 0$ means f is increasing at $x = 2$.

(E) $x = 2$ cannot be an extremum because $f'(2) = 0$.

Solution. $f'(2) = 0$ only tells us $x = 2$ is a critical point, where the tangent is momentarily horizontal. Whether it is a max, a min, or neither depends on how f' *changes sign*: a relative maximum requires f' to go from positive to negative at $x = 2$. Without that sign change the point could be a relative minimum or a shelf (as with $f(x) = x^3$ at $x = 0$). The answer is **(C)**.

Independent Problem Solving

Independent 1: Increasing intervals from a factored derivative

A function f has $f'(x) = (x-1)(x-5)$. On which interval(s) is f increasing?

(A) $(1, 5)$

(B) $(-\infty, 1) \cup (5, \infty)$

(C) $(-\infty, 1)$

(D) $(5, \infty)$

(E) all real numbers

Solution. f increases where $f'(x) > 0$. The product $(x-1)(x-5)$ is positive when both factors share a sign (outside the roots), i.e. on $(-\infty, 1)$ and $(5, \infty)$; it is negative between 1 and 5. The answer is **(B)**.

Independent 2: Counting relative extrema

A function f has $f'(x) = x^2(x+4)$. How many relative extrema does f have?

(A) 0

(B) 1

(C) 2

(D) 3

(E) Cannot be determined.

Solution. Critical points: $x = 0$ and $x = -4$. The factor $x^2 \geq 0$ never changes sign, so the sign of f' follows $(x+4)$. At $x = -4$: f' changes from negative (for $x < -4$) to positive (for $-4 < x < 0$), a relative minimum. At $x = 0$: f' is positive on both sides ($x^2 > 0$ and $x + 4 > 0$), no sign change, so neither. Exactly one relative extremum. The answer is **(B)**.

Independent 3 (FRQ): Full First Derivative Test analysis

Let $f(x) = x^4 - 4x^3$.

(a) Find all critical points.

(b) Find the intervals where f is increasing and decreasing. Justify.

(c) Classify each critical point and give the value of f at any relative extremum.

Solution.

(a) Critical points.

> Differentiate and factor:
> $$f'(x) = 4x^3 - 12x^2 = 4x^2(x-3).$$
> Setting $4x^2(x-3) = 0$ gives $x = 0$ and $x = 3$ (and f' exists everywhere), so those are the critical points.

(b) Where f is increasing and decreasing.

> The factor $4x^2 \geq 0$, so the sign of f' follows $(x-3)$: negative for $x < 0$ and for $0 < x < 3$, positive for $x > 3$. Therefore f is decreasing on $(-\infty, 0)$ and $(0, 3)$, because $f' < 0$ there, and increasing on $(3, \infty)$, because $f' > 0$ there.

(c) Classifying the critical points.

> At $x = 0$, f' does not change sign (negative on both sides), so $x = 0$ is neither. At $x = 3$, f' changes from negative to positive, a relative minimum; its value is $f(3) = 3^4 - 4 \cdot 3^3 = 81 - 108 = -27$. So f has a relative minimum value of -27 at $x = 3$, and the critical point $x = 0$ is not an extremum.

Independent 4 (FRQ): Candidates Test

Find the absolute maximum and minimum values of $f(x) = x^3 - 12x + 1$ on $[-3, 3]$ and where they occur.

Solution. f is continuous on the closed interval $[-3, 3]$, so the Candidates Test applies. Differentiating and factoring,

$$f'(x) = 3x^2 - 12 = 3\big(x^2 - 4\big) = 3(x-2)(x+2),$$

so the critical points are $x = -2$ and $x = 2$, both inside $(-3, 3)$. The candidates are $x = -3, -2, 2, 3$. Evaluating,

$$\begin{aligned} f(-3) &= (-27) - 12(-3) + 1 = -27 + 36 + 1 = 10, \\ f(-2) &= (-8) - 12(-2) + 1 = -8 + 24 + 1 = 17, \\ f(2) &= 8 - 24 + 1 = -15, \\ f(3) &= 27 - 36 + 1 = -8. \end{aligned}$$

Comparing 10, 17, -15, -8: the absolute maximum value is 17, at $x = -2$, and the absolute minimum value is -15, at $x = 2$.

Independent 5: Reading a sign table

The table gives the sign of $f'(x)$ on the indicated intervals for a differentiable function f.

Interval	$(-\infty, -1)$	$(-1, 2)$	$(2, 5)$	$(5, \infty)$
Sign of f'	$-$	$+$	$+$	$-$

At which x-value does f have a relative maximum?

(A) $x = -1$

(B) $x = 2$

(C) $x = 5$

(D) $x = -1$ and $x = 5$

(E) There is no relative maximum.

Solution. A relative maximum occurs where f' changes from $+$ to $-$. Scanning the signs $-, +, +, -$: at $x = -1$ it goes $-$ to $+$ (relative min); at $x = 2$ it stays $+$ (no change); at $x = 5$ it goes $+$ to $-$ (relative max). The answer is **(C)**.

Independent 6: Classifying a cusp

Let $f(x) = x^{2/3}$. What occurs at $x = 0$?

(A) Relative maximum

(B) Relative minimum

(C) Neither (f' does not change sign)

(D) $x = 0$ is not in the domain of f

(E) $f'(0) = 0$

Solution. With $f(x) = x^{2/3}$, differentiate using the power rule:

$$f'(x) = \frac{2}{3}x^{-1/3} = \frac{2}{3x^{1/3}}.$$

This does not exist at $x = 0$, while $x = 0$ *is* in the domain ($f(0) = 0$), so $x = 0$ is a critical point. For $x < 0$, $x^{1/3} < 0 \Rightarrow f' < 0$; for $x > 0$, $f' > 0$. Since f' changes from negative to positive, f has a relative minimum at $x = 0$. The answer is **(B)**.

Independent 7 (FRQ): Candidates Test with trigonometric values

Find the absolute maximum and minimum values of $f(x) = \sin x + \cos x$ on $[0, 2\pi]$ and where they occur.

Solution. f is continuous on the closed interval $[0, 2\pi]$, so the Candidates Test applies. Differentiating $f(x) = \sin x + \cos x$,

$$f'(x) = \cos x - \sin x.$$

Setting $f'(x) = 0$ gives $\cos x = \sin x$, i.e. $\tan x = 1$, so on $[0, 2\pi]$ the critical points are $x = \frac{\pi}{4}$ and $x = \frac{5\pi}{4}$. The candidates are $x = 0,\ \frac{\pi}{4},\ \frac{5\pi}{4},\ 2\pi$. Evaluating,

$$\begin{aligned} f(0) &= \sin 0 + \cos 0 = 0 + 1 = 1, \\ f\left(\tfrac{\pi}{4}\right) &= \tfrac{\sqrt{2}}{2} + \tfrac{\sqrt{2}}{2} = \sqrt{2} \approx 1.414, \\ f\left(\tfrac{5\pi}{4}\right) &= -\tfrac{\sqrt{2}}{2} - \tfrac{\sqrt{2}}{2} = -\sqrt{2} \approx -1.414, \\ f(2\pi) &= \sin 2\pi + \cos 2\pi = 0 + 1 = 1. \end{aligned}$$

Comparing the outputs, the absolute maximum value is $\sqrt{2}$, at $x = \frac{\pi}{4}$, and the absolute minimum value is $-\sqrt{2}$, at $x = \frac{5\pi}{4}$.

AP Calculus AB Module 5B
First Derivative Analysis Overview

This module uses one fact in three ways: the *sign of f'* tells you whether f is rising or falling, and the *sign changes of f'* locate the peaks and valleys. Critical points are the hinges where behavior can switch.

Key facts

$$f'(x) > 0 \Rightarrow f \text{ increasing} \qquad f'(x) < 0 \Rightarrow f \text{ decreasing}$$

Critical point: $f'(c) = 0$ or $f'(c)$ does not exist (and c in the domain).

A critical point is a max/min only if f' actually *changes sign* there.

Building a sign chart
Factor f'; mark every critical point on a number line; test one point per interval in the *factored* f'; label each interval $\nearrow$ ($f' > 0$) or $\searrow$ ($f' < 0$).

First Derivative Test (relative extrema, anywhere)

1. If f' changes from positive to negative at c: relative **maximum** at $x = c$.
2. If f' changes from negative to positive at c: relative **minimum** at $x = c$.
3. If f' does not change sign at c: **neither**.

Candidates Test (absolute extrema, closed interval $[a, b]$, f continuous)
Evaluate f at every critical point in (a, b) *and* at both endpoints; the largest f-value is the absolute max, the smallest is the absolute min. Compare y-values, and never forget the endpoints.

Justifying (the language readers reward)

"f is *(increasing/decreasing)* on *(interval)* because f' is *(positive/negative)* there."

"f has a relative *(maximum/minimum)* at $x = c$ because f' changes from *(positive to negative / negative to positive)* there."

"The absolute *(max/min)* is *(value)* at $x =$*(location)*, by comparing f at the critical points and endpoints."

Common traps
(1) $f'(c) = 0$ does *not* guarantee an extremum; check the sign change. (2) Don't forget critical points where f' *does not exist* (corners/cusps). (3) Relative extrema can't occur at endpoints, but absolute extrema can. (4) On a graph of f', read its *height* (sign), not the direction it slopes: "f' increasing" $\neq$ "f increasing."